Report

Current Biology

Evidence for a Mid-Jurassic Adaptive Radiation in Mammals

Highlights

- Rates of morphological evolution and disparity are quantified in Mesozoic mammals
- Elevated rates prior to the Late Jurassic support a mid-Jurassic adaptive radiation
- Slower rates characterized the Late Jurassic to the end-Cretaceous
- Early evolution in Theria was exceptionally rapid, but then slowed dramatically

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In Brief

Close et al. show high rates of morphological evolution and disparity in Mesozoic mammals during the Early to Middle Jurassic—evidence of a major adaptive radiation. Rates were generally low following this interval. Morphological rates at the origin of Theria were especially high, but subsequently remained low, despite taxonomic diversification.

Close et al., 2015, Current Biology *25*, 2137–2142
August 17, 2015 ©2015 Elsevier Ltd All rights reserved
http://dx.doi.org/10.1016/j.cub.2015.06.047

Current Biology
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CellPress

Evidence for a Mid-Jurassic Adaptive Radiation in Mammals

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http://dx.doi.org/10.1016/j.cub.2015.06.047

SUMMARY

A series of spectacular discoveries have transformed our understanding of Mesozoic mammals in recent years. These finds reveal hitherto-unsuspected eco-morphological diversity that suggests that mammals experienced a major adaptive radiation during the Middle to Late Jurassic [1]. Patterns of mammalian macroevolution must be reinterpreted in light of these new discoveries [1–3], but only taxonomic diversity and limited aspects of morphological disparity have been quantified [4, 5]. We assess rates of morphological evolution and temporal patterns of disparity using large datasets of discrete characters. Rates of morphological evolution were significantly elevated prior to the Late Jurassic, with a pronounced peak occurring during the Early to Middle Jurassic. This intense burst of phenotypic innovation coincided with a stepwise increase in apparent long-term standing diversity [4] and the attainment of maximum disparity, supporting a "short-fuse" model of early mammalian diversification [2, 3]. Rates then declined sharply, and remained significantly low until the end of the Mesozoic, even among therians. This supports the "long-fuse" model of diversification in Mesozoic therians. Our findings demonstrate that sustained morphological innovation in Triassic stem-group mammals culminated in a global adaptive radiation of crown-group members during the Early to Middle Jurassic.

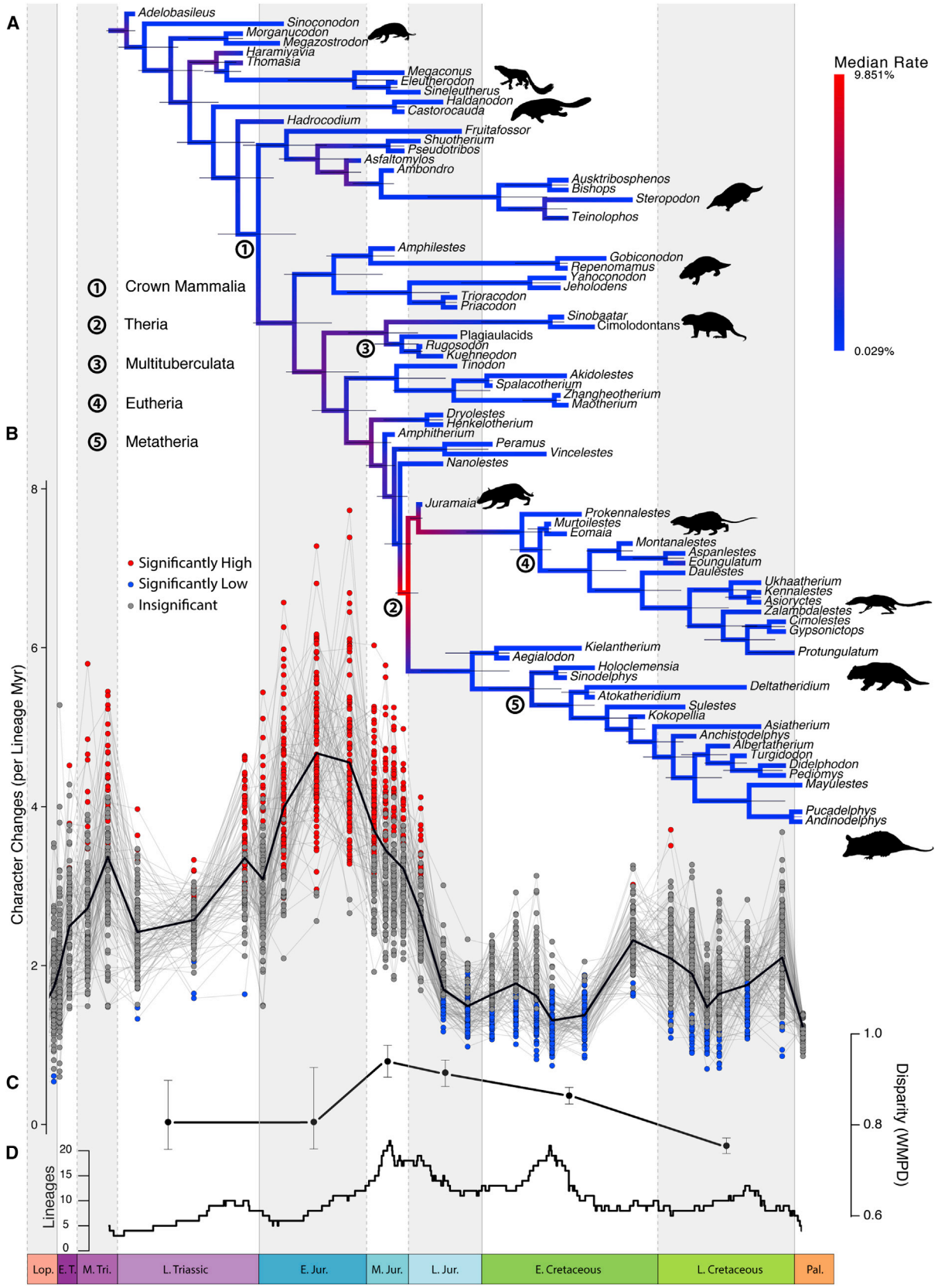
RESULTS

Quantifying rates of morphological evolution is crucial for interpreting macroevolutionary events in deep time, especially when assessing potential adaptive radiations [6–8]. Contrary to the traditional view that Mesozoic mammals were exclusively small, generalized insectivores [9, 10], discoveries in the last two decades, especially from China, have demonstrated that they were adapted for diverse feeding and locomotor ecologies [11]. These finds extend the early mammal repertoire to include digging [12, 13], climbing [14], gliding [15], and swimming [16, 17] and show that some non-therian lineages achieved surprisingly large body sizes (up to approximately 1 kg [18]). Much of this morphological diversity appeared rapidly in a Middle Jurassic "wave of diversification," first recognized by Luo [2, 3, 19] and interpreted as a major adaptive radiation by Meng [1]. The hypothesis of adaptive radiation makes several quantitatively testable predictions, including the occurrence of high early rates of lineage diversification and phenotypic evolution, leading to substantial increases in taxonomic diversity and morphological disparity [8]. Rigorous statistical analyses have demonstrated substantial increases in mammal diversity during the Middle Jurassic [4]. However, co-occurring rates of phenotypic evolution and their relation to observed patterns of disparity have not been explored using quantitative analyses.

Rates of Morphological Evolution

Few studies quantify patterns of morphological evolution in Mesozoic mammals, and none rigorously analyze rates of morphological evolution across more than two species [20]. Across Mesozoic mammals as a whole, patterns of disparity in feeding ecomorphology have been quantified via geometric morphometric analysis of mandibular outline morphology coupled with linear measurements and qualitative categorization of tooth morphology [5], while linear measurements have been used to characterize locomotor ecology [11]. Analyses of dental complexity and body size have thrown light on the Late Cretaceous diversification of multituberculates [21]. Large datasets of discrete characters, by contrast, can potentially provide a more comprehensive overview of the morphological transformations that occur during major evolutionary transitions, as they document anatomical changes occurring throughout the skeleton, including many comparisons not amenable to continuous measurements [6, 22, 23].

Several recent studies evaluated rates of morphological evolution by co-opting rich datasets of discrete characters originally constructed for phylogenetic inference [6, 20, 22–28]. This emerging technique has yielded stimulating results and could significantly advance our understanding of macroevolution. However, little consideration has thus far been given to potential limitations of this approach. Two key biases may affect estimates of rates or disparity based on morphological datasets constructed for phylogenetics: (1) unrepresentative inclusion of taxa and (2) unrepresentative formulation or inclusion of characters. Either bias could result from researcher-specific choices introduced during dataset construction, from community-wide historical research efforts (e.g., an emphasis on documenting the assembly of therian dentitions), or from biased availability of specimens and taxa in the fossil record. We countered these



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biases using a range of sensitivity analyses, including taxonomic jackknifing procedures and the investigation of alternative source datasets, including those focused on non-therian mammals (see the Supplemental Information). Scaling of branch durations in trees can also impact rate estimates. We addressed this by using multiple tree-scaling methods bracketing a wide range of divergence times for key mammalian nodes.

Our principal rates and disparity analyses were performed on the morphological character matrix of Zhou et al. [29], but sensitivity analyses performed on several other datasets show that these results are representative (see the Supplemental Information). Performing separate significance tests for rates along internal and terminal branches (DRYAD Figure S7) did not substantively affect the observed patterns, suggesting that our results are not biased by the exclusion of autapomorphies (a unique feature of terminal branches that are rarely sampled by phylogenetic character matrices).

Mammals were characterized by significantly elevated rates of morphological evolution for the first one-third to one-half of their history under all iterations of our analyses (Figure 1). This was found both using the full dataset and using only dental characters and whether or not the divergence times of phylogenetic nodes were estimated using morphological clocks (based on the posterior tree sample generated by BEAST 2; e.g., [30]) or other algorithmic time-scaling methods applied to randomly sampled most-parsimonious trees (MPTs) (see the Supplemental Results). The same patterns were also recovered by sensitivity analyses that excluded highly complete (Lagerstätten) taxa and that jackknifed the taxonomic dataset via repeated, random subsampling without replacement.

Evolutionary rates were estimated using BEAST 2 and the R package Claddis [31]. In Claddis, ancestral character states were first estimated via maximum-likelihood methods, allowing evolutionary rates for each branch of the time-scaled tree to be calculated based on the total number of inferred character changes, corrected for missing data. Likelihood-ratio tests were then used to identify branches with significantly higher or lower rates than the pooled rate for the rest of the tree. Our main-text figures show results for the full dataset, excluding non-mammalian cynodonts, using the posterior sample of trees obtained from a Bayesian morphological clock analysis performed with BEAST 2 (for results including non-mammalian cynodonts, see the Supplemental Results and DRYAD Figures S2–S6; for dental characters only, see DRYAD Figures S40–S42). This method provides the greatest congruence with molecular clock estimates of crown-mammalian and therian divergence times [32] and integrates topological uncertainty (Table S1; DRYAD Figures S34–S37). Maximum rates of phenotypic evolution occurred near the Early/Middle Jurassic boundary (“mid-Jurassic”). Subsequently, in the early Late Jurassic, rates declined and remained significantly low until the end of the Mesozoic. Small but statistically insignificant peaks occur during the Cretaceous. High Early to Middle Jurassic rates do not solely reflect evolution of the therian stem group, as elevated rates also occur in australosphenidans (the monotreme stem group) and multituberculates (Figure 1A). Critically, analysis of the matrices of Yuan et al. [33] and Krause et al. [34], which primarily document the evolution of allotherians, including multituberculates (clades that are poorly sampled in our focal matrix), confirms that high Early to Late Jurassic rates of evolution were not confined solely to the therian stem lineage (DRYAD Figures S12–S17).

Elevated rates could be an artifact of the density of sampling of phylogenetic lineages. Phylogenetic lineage diversity counted directly from the phylogeny (i.e., the sum of observed taxa and ghost lineages within intervals) shows two peaks, one in the Middle Jurassic and another in the Early Cretaceous (Figure 1D). The Early Cretaceous peak is not associated with high rates, indicating that although Middle Jurassic mammals are proportionally over-sampled in the phylogenetic dataset, this is unlikely to have inflated our estimates of their evolutionary rates.

Multituberculates underwent a substantial Late Cretaceous radiation as measured by dental complexity [21] but are insufficiently sampled in our focal matrix. Therefore, we also analyzed one dataset that explicitly targets multituberculates and another that targets allotherians more generally (those of Yuan et al. [33] and Krause et al. [34], respectively; see the Supplemental Results). These analyses did not find consistent statistical support for elevated rates in Cretaceous multituberculates and recovered essentially static levels of disparity from the Late Jurassic to the Eocene.

Most internal branches along the backbone of the tree between crown Mammalia and Boreosphenida show significantly elevated rates of phenotypic evolution. However, Theria only saw significantly fast rates of evolution during its origin in the Middle Jurassic, with greater support for fast rates at the root of Eutheria than at the root of Theria. When rates are inferred simultaneously with phylogeny and divergence times using a Bayesian morphological clock approach, the fastest rates on the tree are found to occur along the branch leading to Theria (~12.8 times faster than the mean rate, calculated from the MCC tree). The second fastest rate occurs along the branch

Figure 1. Morphological Rates, Disparity, and Phylogenetic Lineage Diversity in Mesozoic Mammals

(A) Maximum clade credibility (MCC) tree calculated from the posterior tree sample obtained via a Bayesian morphological clock analysis of the complete morphological character dataset of Zhou et al. [29] for Mesozoic and extant mammals (non-mammalian cynodonts and extant taxa not shown), using BEAST 2. Encircled numbers denote key clades. The heatmap shows median rates of phenotypic evolution (percent character change per Ma) estimated simultaneously with topology and branch lengths (red indicates fast rates, blue slow). Node bars indicate 95% highest-probability densities for node divergence times. Silhouettes are from http://www.phylopic.org.

(B) Time-series “spaghetti” plot showing significantly fast (red) or slow (blue) rates of phenotypic evolution, inferred with Claddis using maximum-likelihood methods via analysis of 100 trees from the Bayesian posterior sample generated by BEAST 2 (age-level time bins). Gray points represent bins for which rates were inferred to be insignificantly different to the pooled average. Each line represents the analysis of a single chronogram from the Bayesian posterior tree sample. Rates prior to the Triassic have been omitted as the data were deemed insufficient for reliable estimates.

(C) Weighted mean pairwise disparity (WMPD) estimated for epoch-level bins from the discrete morphological character data. Error bars represent 95% confidence intervals.

(D) Phylogenetic lineage diversity counted directly from the phylogeny (i.e., the sum of observed taxa and ghost lineages within intervals).

leading to Eutheria (~8.7 times faster than the mean). Both of these branches are short, which increases the likelihood of inferring fast rates. However, branches leading to successive outgroups of Theria are similarly short but are not characterized by especially elevated rates. Conversely, the slowest rates inferred are ~28.2 times slower than the mean. Fast rates on the branches leading to Theria and Eutheria could result from a systematic bias toward documenting the evolutionary assembly of therian dentitions and anatomy by previous workers. However, this potential bias is not driving the overall pattern of high early rates as the ages of basal therian and eutherian nodes are younger than the Early/Middle Jurassic peak in mammalian time series rates (Figure 1).

Disparity, quantified from the discrete character data as the within-interval weighted mean pairwise dissimilarity, increased through the Late Triassic and Early Jurassic. Maximum disparity was attained in the Middle Jurassic, coincident with maximum rates of evolution, and subsequently declined through the Upper Jurassic, Early Cretaceous, and into the Late Cretaceous. However, disparity is likely to be underestimated for the Late Cretaceous due to the omission of multituberculates from this interval (see below).

DISCUSSION

An intense burst of morphological innovation among Early to Middle Jurassic mammals was integral to the origin of crown-group Mammalia and resulted in the rapid appearance of anatomically distinctive order- or family-level lineages. These include eutriconodontans, multituberculates, docodontans, shuotheriids, australosphenidans, amphitheriids, and other cladotherian groups such as dryolestoids [35], metatherians, and eutherians [3]. This burst not only generated high levels of ecomorphological diversity [1–3, 5], but also coincided with the acquisition of key mammalian features of the dentition, middle ear, and shoulder girdle [2, 3, 19, 36]. This pattern is consistent with the "short-fuse" model of Middle Jurassic diversification proposed by Luo [2, 3] (usage of the term referring to a combination of ecomorphological and taxonomic diversification) and closely mirrors the bursts in rates associated with major shifts in ecomorphology observed in post-Palaeozoic echinoids [23], suggesting that such bursts may be commonly associated with the exploration of ecospace in novel adaptive zones. Morphological disparity, based on discrete anatomical characters, also peaked in the Middle Jurassic, coinciding with a peak in phylogenetic lineage diversity, but preceding the Late Jurassic peaks in mandibular/dental disparity [5] and subsampled taxonomic diversity [4] estimated from the fossil record. This disjunction is most likely due to the temporal extension of ghost ranges for the diverse mid- to Late Jurassic fauna by our stratigraphic calibration methods.

Although evolutionary rates fell sharply at the beginning of the Late Jurassic, standing diversity [4] and mandibular/dental disparity [5] remained high until the mid-Cretaceous (an interval that saw a loss of dental diversity that was not immediately replaced); later peaks in standing diversity and mandibular/dental disparity are not associated with elevated rates of morphological evolution in our analyses. Notably, the second peak in phylogenetic lineage diversity during the Early Cretaceous coincides with significantly slow rates. This suggests that ecomorphological evolution and taxonomic diversification, which were closely coupled during the mid-Jurassic radiation, were decoupled during subsequent evolution. Decoupled rates of morphological evolution and taxonomic diversification have also been observed in the later stages of Cenozoic mammal radiations [37].

The tribosphenic molar is frequently cited as a key innovation of therian mammals [2, 38, 39]. The occurrence of exceptional rates of morphological evolution at the deepest therian divergences is consistent with this hypothesis. Nevertheless, slow morphological and body-size evolution among post-mid-Jurassic therians is consistent with qualitative interpretations suggesting a "long fuse" of therian diversification [1, 3]. The disparity analysis of Grossnickle and Polly [5] also found that therians did not experience a substantial radiation until the Cretaceous Terrestrial Revolution (KTR).

The early-burst pattern of Mesozoic mammalian evolution (perhaps more accurately described as a "delayed early burst"; cf. Hopkins and Smith [23]) closely mirrors patterns associated with the origins of other major vertebrate groups, including lungfishes [6], tetrapods [27], amniotes [40], archosaurs [28], and birds [30, 41]. These patterns are consistent with the occurrence of large-scale adaptive radiation (sensu Osborn [42] and Simpson [43]), as is the observation of parallel, evolutionarily independent invasions of ecological niches [8]. Independent acquisition of similar dental and cranial characters by multiple mammalian lineages occurred during the Early/Middle Jurassic, including the convergent origins of mortar-and-pestle dental types in australosphenidans, shuotheriids, and boreosphenidans; of multifunctional teeth in docodontans; and of the inner-ear cochlear coiling in cladotherians and monotremes [44]. Rates of morphological innovation were not significantly elevated again until the radiation of placentals in the early Cenozoic.

Explanations for adaptive radiation in mid-Jurassic mammals remain conjectural. Adaptive radiations are driven by ecological opportunity due to the invasion of new habitat, the extinction of incumbents, or the acquisition of a key innovation [45]. One possibility is that this burst of morphological evolution was tied to the breakup of Pangaea, which began in the Early to Middle Jurassic; another is that it was initiated by the acquisition of a "critical mass" of key anatomical or physiological innovations, which had been steadily accruing since the origin of the mammalian lineage and earlier, in the non-mammalian cynodonts. Whatever the cause, this burst of rapid evolution in the mid-Jurassic burst resulted in the origin of most of the major groups that would go on to dominate mammalian diversity through the remainder of the Mesozoic.

EXPERIMENTAL PROCEDURES

Morphological Dataset and Phylogenetic Analyses

For simplicity, we use the term "mammal," rather than "mammaliaform," to refer to the common ancestor of *Sinoconodon*, living mammals, and all its descendants. Primary analyses were performed on the morphological character matrix compiled by Zhou et al. [29], comprising 110 taxa and 475 characters, including Mesozoic, Cenozoic, and extant taxa. Although this dataset represents the most thoroughly sampled matrix in terms of taxonomic coverage, additional morphological character matrices were used to determine sensitivity to taxonomic and character composition (see the Supplemental Experimental Procedures).

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Primary analyses were run on the posterior tree sample obtained from a Bayesian morphological clock analysis of the Zhou et al. [29] dataset using BEAST 2.1.3 [46], rooted on the non-mammalian cynodont *Thrinaxodon*. XML input files for BEAST 2 were generated using the R package BEASTmasteR [47]. The BEAST 2 analysis was repeated four times to ensure that stationarity (convergence) had been reached. In addition to morphological character data, stratigraphic information was provided in the form of tip ages for terminal taxa. The Mk model of morphological character evolution for unordered characters was used. A four-category gamma distribution for among-site rate variation was used, and the BDSKY (Birth Death Skyline) tree model was specified. A random starting tree was used. A root prior was not specified. The MCC tree was calculated, which includes median estimates of rates of character evolution. Additional sensitivity analyses (see the Supplemental Experimental Procedures) were performed on sets of algorithmically scaled MPTs from parsimony analyses using PAUP* 4.0b10 [48]. Post-Mesozoic taxa were dropped from all trees and morphological datasets prior to conduction of further analyses.

Rates of Morphological Evolution

Rates of morphological evolution were inferred using two methods: (1) in BEAST 2, simultaneously with topology and branch lengths/divergence dates, and (2) with maximum-likelihood methods on 100 randomly selected trees drawn from the posterior sample obtained from BEAST 2 using the function DiscreteCharacterRate in R package Claddis [31]. Claddis reconstructs ancestral character states using maximum likelihood and identifies branches with significantly higher or lower rates than the pooled rate for the rest of the tree via likelihood-ratio tests. Time-series "spaghetti" plots were generated to show the temporal variation in rates, binned by stage. Claddis routines were also performed on randomly drawn MPTs, time-scaled algorithmically using the minimum branch length (MBL) method (see the Supplemental Experimental Procedures).

Disparity Calculations

Morphological disparity curves were calculated from the discrete morphological character data for epoch-level time bins using WMPD. First, the maximum observable distance (MOD) dissimilarity matrix was calculated in R using the Claddis function MorphDistMatrix. Within-bin WMPD was then calculated as:

$$\frac{\sum \boldsymbol{D} \times \boldsymbol{C}}{\sum \boldsymbol{C}},$$

where $\boldsymbol{D}$ is the upper triangle of the matrix of pairwise dissimilarities and $\boldsymbol{C}$ is the upper triangle of the matrix of comparable characters. WMPD thus places greater importance on dissimilarities based on many comparable characters. WMPD is robust to sample-size variation [49] and is superior to methods that involve ordination of the character data (e.g., with principal coordinates analysis) and calculation of multivariate variance, as taxa lacking any comparable characters need not be deleted [50, 51]. Bootstrapped 95% confidence intervals were calculated for WMPD.

ACCESSION NUMBERS

The BEAST 2 XML file and R scripts for all analyses performed in this study, together with all input data and output files, are available on DRYAD: http://doi.org/10.5061/dryad.760sc.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Results, Supplemental Experimental Procedures, and one table and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2015.06.047.

AUTHOR CONTRIBUTIONS

Conceptualization, R.B.J.B. and R.A.C.; Methodology, R.A.C., R.B.J.B., G.T.L., and M.F.; Software, R.A.C. and G.T.L.; Formal Analysis, R.A.C.; Investigation, R.A.C. and R.B.J.B.; Data Curation, R.A.C.; Writing – Original Draft, R.A.C.; Writing – Review & Editing, R.A.C., R.B.J.B., M.F., and G.T.L.; Visualization, R.A.C.; Supervision, R.B.J.B. and M.F.; Funding Acquisition, M.F.

ACKNOWLEDGMENTS

We wish to thank the Leverhulme Trust (RPG-2012-658) for support, Nick Matzke for assistance with implementing BEASTmasteR; the editor, Florian Maderspacher; and three anonymous reviewers for providing comments that greatly improved the manuscript. G.T.L. is supported by Australian Research Council grant DE140101879.

Received: March 31, 2015
Revised: May 13, 2015
Accepted: June 18, 2015
Published: July 16, 2015

REFERENCES

1. Meng, J. (2014). Mesozoic mammals of China: implications for phylogeny and early evolution of mammals. Natl. Sci. Rev. *1*, 521–542.
2. Luo, Z.-X. (2007). Transformation and diversification in early mammal evolution. Nature *450*, 1011–1019.
3. Luo, Z.X. (2007). Successive diversifications in early mammalian evolution. In Major Transitions in Vertebrate Evolution, J.S. Anderson, and H.-D. Sues, eds. (Bloomington: Indiana University Press), pp. 337–391.
4. Newham, E., Benson, R., Upchurch, P., and Goswami, A. (2014). Mesozoic mammaliaform diversity: The effect of sampling corrections on reconstructions of evolutionary dynamics. Palaeogeogr. Palaeocl. *412*, 32–44.
5. Grossnickle, D.M., and Polly, P.D. (2013). Mammal disparity decreases during the Cretaceous angiosperm radiation. Proc. Biol. Sci. *280*, 20132110–20132110.
6. Lloyd, G.T., Wang, S.C., and Brusatte, S.L. (2012). Identifying heterogeneity in rates of morphological evolution: discrete character change in the evolution of lungfish (Sarcopterygii; Dipnoi). Evolution *66*, 330–348.
7. Schluter, D. (2000). The Ecology of Adaptive Radiation. (Oxford: Oxford University Press).
8. Gavrilets, S., and Losos, J.B. (2009). Adaptive radiation: contrasting theory with data. Science *323*, 732–737.
9. Simpson, G.G. (1929). American Mesozoic Mammalia. (Newhaven: Yale University Press).
10. Lillegraven, J.A., Kielan-Jaworowska, Z., and Clemens, W.A. (1979). Mesozoic Mammals. (Berkeley: University of California Press).
11. Chen, M., and Wilson, G.P. (2015). A multivariate approach to infer locomotor modes in Mesozoic mammals. Paleobiology *41*, 1–33.
12. Luo, Z.-X., Chen, P., Li, G., and Chen, M. (2007). A new eutriconodont mammal and evolutionary development in early mammals. Nature *446*, 288–293.
13. Luo, Z.-X., Meng, Q.-J., Ji, Q., Liu, D., Zhang, Y.-G., and Neander, A.I. (2015). Mammalian evolution. Evolutionary development in basal mammaliaforms as revealed by a docodontan. Science *347*, 760–764.
14. Meng, Q.-J., Ji, Q., Zhang, Y.-G., Liu, D., Grossnickle, D.M., and Luo, Z.-X. (2015). Mammalian evolution. An arboreal docodont from the Jurassic and mammaliaform ecological diversification. Science *347*, 764–768.
15. Meng, J., Hu, Y., Wang, Y., Wang, X., and Li, C. (2006). A Mesozoic gliding mammal from northeastern China. Nature *444*, 889–893.
16. Ji, Q., Luo, Z.-X., Yuan, C.-X., and Tabrum, A.R. (2006). A swimming mammaliaform from the Middle Jurassic and ecomorphological diversification of early mammals. Science *311*, 1123–1127.
17. Martin, T. (2005). Postcranial anatomy of *Haldanodon exspectatus* (Mammalia, Docodonta) from the Late Jurassic (Kimmeridgian) of Portugal and its bearing for mammalian evolution. Zool. J. Linn. Soc. *145*, 219–248.

18. Hu, Y., Meng, J., Wang, Y., and Li, C. (2005). Large Mesozoic mammals fed on young dinosaurs. Nature *433*, 149–152.
19. Kielan-Jaworowska, Z., Cifelli, R.L., and Luo, Z.-X. (2004). Mammals from the Age of Dinosaurs. (New York: Columbia University Press).
20. Ruta, M., Botha-Brink, J., Mitchell, S.A., and Benton, M.J. (2013). The radiation of cynodonts and the ground plan of mammalian morphological diversity. Proc. Biol. Sci. *280*, 20131865.
21. Wilson, G.P., Evans, A.R., Corfe, I.J., Smits, P.D., Fortelius, M., and Jernvall, J. (2012). Adaptive radiation of multituberculate mammals before the extinction of dinosaurs. Nature *483*, 457–460.
22. Brusatte, S.L., Lloyd, G.T., Wang, S.C., and Norell, M.A. (2014). Gradual assembly of avian body plan culminated in rapid rates of evolution across the dinosaur-bird transition. Curr. Biol. *24*, 2386–2392.
23. Hopkins, M.J., and Smith, A.B. (2015). Dynamic evolutionary change in post-Paleozoic echinoids and the importance of scale when interpreting changes in rates of evolution. Proc. Natl. Acad. Sci. USA *112*, 3758–3763.
24. Beck, R.M.D., and Lee, M.S.Y. (2014). Ancient dates or accelerated rates? Morphological clocks and the antiquity of placental mammals. Proc. Biol. Sci. *281*, 20141278–20141278.
25. Lee, M.S.Y., Soubrier, J., and Edgecombe, G.D. (2013). Rates of phenotypic and genomic evolution during the Cambrian explosion. Curr. Biol. *23*, 1889–1895.
26. Lee, M.S.Y., Cau, A., Naish, D., and Dyke, G.J. (2014). Morphological clocks in paleontology, and a mid-Cretaceous origin of crown Aves. Syst. Biol. *63*, 442–449.
27. Ruta, M., Wagner, P.J., and Coates, M.I. (2006). Evolutionary patterns in early tetrapods. I. Rapid initial diversification followed by decrease in rates of character change. Proc. Biol. Sci. *273*, 2107–2111.
28. Brusatte, S.L., Benton, M.J., Ruta, M., and Lloyd, G.T. (2008). Superiority, competition, and opportunism in the evolutionary radiation of dinosaurs. Science *321*, 1485–1488.
29. Zhou, C.-F., Wu, S., Martin, T., and Luo, Z.-X. (2013). A Jurassic mammaliaform and the earliest mammalian evolutionary adaptations. Nature *500*, 163–167.
30. Lee, M.S.Y., Cau, A., Naish, D., and Dyke, G.J. (2014). Dinosaur evolution. Sustained miniaturization and anatomical innovation in the dinosaurian ancestors of birds. Science *345*, 562–566.
31. Lloyd, G. T. (2015). Claddis: Measuring Morphological Diversity and Evolutionary Tempo. R Package Version 0.1. https://github.com/graemetlloyd/Claddis.
32. dos Reis, M., Donoghue, P.C., and Yang, Z. (2014). Neither phylogenomic nor palaeontological data support a Palaeogene origin of placental mammals. Biol. Lett. *10*, 20131003.
33. Yuan, C.-X., Ji, Q., Meng, Q.-J., Tabrum, A.R., and Luo, Z.-X. (2013). Earliest evolution of multituberculate mammals revealed by a new Jurassic fossil. Science *341*, 779–783.
34. Krause, D.W., Hoffmann, S., Wible, J.R., Kirk, E.C., Schultz, J.A., von Koenigswald, W., Groenke, J.R., Rossie, J.B., O'Connor, P.M., Seiffert, E.R., et al. (2014). First cranial remains of a gondwanatherian mammal reveal remarkable mosaicism. Nature *515*, 512–517.
35. Averianov, A.O., Martin, T., and Lopatin, A. (2014). The oldest dryolestid mammal from the Middle Jurassic of Siberia. J. Vertebrate Paleontol. *34*, 924–931.
36. Luo, Z.X., Crompton, A.W., and Sun, A.L. (2001). A new mammaliaform from the early Jurassic and evolution of mammalian characteristics. Science *292*, 1535–1540.
37. Raia, P., Carotenuto, F., Passaro, F., Piras, P., Fulgione, D., Werdelin, L., Saarinen, J., and Fortelius, M. (2013). Rapid action in the Palaeogene, the relationship between phenotypic and taxonomic diversification in Coenozoic mammals. Proc. Biol. Sci. *280*, 20122244.
38. Woodburne, M.O., Rich, T.H., and Springer, M.S. (2003). The evolution of tribospheny and the antiquity of mammalian clades. Mol. Phylogenet. Evol. *28*, 360–385.
39. Schultz, J.A., and Martin, T. (2014). Function of pretribosphenic and tribosphenic mammalian molars inferred from 3D animation. Naturwissenschaften *101*, 771–781.
40. Anderson, P.S.L., Friedman, M., and Ruta, M. (2013). Late to the table: diversification of tetrapod mandibular biomechanics lagged behind the evolution of terrestriality. Integr. Comp. Biol. *53*, 197–208.
41. Benson, R.B.J., Campione, N.E., Carrano, M.T., Mannion, P.D., Sullivan, C., Upchurch, P., and Evans, D.C. (2014). Rates of dinosaur body mass evolution indicate 170 million years of sustained ecological innovation on the avian stem lineage. PLoS Biol. *12*, e1001853.
42. Osborn, H.F. (1902). The law of adaptive radiation. Am. Nat. *36*, 353–363.
43. Simpson, G.G. (1953). The Major Features of Evolution. (New York: Columbia University Press).
44. Luo, Z.-X., Ruf, I., Schultz, J.A., and Martin, T. (2011). Fossil evidence on evolution of inner ear cochlea in Jurassic mammals. Proc. Biol. Sci. *278*, 28–34.
45. Wellborn, G.A., and Langerhans, R.B. (2015). Ecological opportunity and the adaptive diversification of lineages. Ecol. Evol. *5*, 176–195.
46. Bouckaert, R., Heled, J., Kühnert, D., Vaughan, T., Wu, C.-H., Xie, D., Suchard, M.A., Rambaut, A., and Drummond, A.J. (2014). BEAST 2: a software platform for Bayesian evolutionary analysis. PLoS Comput. Biol. *10*, e1003537.
47. Matzke, N.J. (2014). BEASTmasteR: Automated Conversion of NEXUS Data to BEAST2 XML Format, for Fossil Tip-Dating and Other Uses. http://phylo.wikidot.com/beastmaster.
48. Swofford, D.L. (2002). PAUP*. Phylogenetic Analysis Using Parsimony (and Other Methods), *Version 4*. (Sunderland: Sinauer Associates).
49. Ciampaglio, C.N., Kemp, M., and McShea, D.W. (2001). Detecting changes in morphospace occupation patterns in the fossil record: characterization and analysis of measures of disparity. Paleobiology *27*, 695–715.
50. Butler, R.J., Brusatte, S.L., Andres, B., and Benson, R.B.J. (2012). How do geological sampling biases affect studies of morphological evolution in deep time? A case study of pterosaur (Reptilia: Archosauria) disparity. Evolution *66*, 147–162.
51. Benson, R.B.J., and Druckenmiller, P.S. (2014). Faunal turnover of marine tetrapods during the Jurassic-Cretaceous transition. Biol. Rev. Camb. Philos. Soc. *89*, 1–23.

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